A Contextual Draft Helper for MTG Arena Powered Cube: Win-Rate Priors, Pool Attention, and Personal Game Feedback

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Abstract

Drafting in Magic: The Gathering requires balancing individual card strength against interactions with previous selections. We present a Powered Cube draft assistant that combines an empirical card-quality prior with candidate-to-pool attention and conditional selection associations. The prior uses 17Lands Games in Hand win rates; learned corrections incorporate card attributes, rules text, and draft context. Personal adaptation combines outcome-weighted pick imitation with an auxiliary game-result objective using the player's deck and observed opponent cards. The model is trained on 87,999 public draft decisions and adapted using 180 personal picks and 67 games. It matches 55.57% of choices in a reused historical validation subset and follows the expected direction in seven of nine contextual diagnostics, with remaining failures involving reanimation targets. In a retrospective single-player study, six drafts without the helper produced 17 wins in 35 games (48.6%); seven drafts associated with the final model produced 31 wins in 48 games (64.6%). Differences in opponents, player decisions, and recommendation adherence prevent a causal interpretation. The results demonstrate contextual score responses and describe observed gameplay outcomes, while leaving the effect of training-data volume unresolved.

1. Introduction

Magic: The Gathering separates two strategic problems: building a deck and playing it. In a draft, players acquire the cards for their decks through a sequence of choices. Eight players each open a 15-card pack, keep one card, and pass the remainder to a neighbor. Selection and passing continue until the packs are empty. Three packs per player yield a pool of 45 cards, from which each player builds a deck of at least 40 cards, adding basic lands as needed (Bertram et al., 2021).

Lands supply mana, the resource used to cast spells. Most spells require particular colors of mana, so powerful cards may be difficult to use together. The drafted pool also exceeds the number of cards needed for the deck, allowing speculative choices and changes of direction. A pick therefore depends on both the available alternatives and the cards already selected (Bertram et al., 2021).

A Cube is a curated collection from which draft packs are assembled. In the Arena Powered Cube studied here, the collection supports powerful interactions alongside individually strong cards. The drafting problem is consequently to assemble a coherent strategy while securing the resources needed to play it.

These objectives can conflict. A reanimation strategy, for example, needs large creatures, ways to place them in the graveyard, and spells that return them to play. Selecting another large creature may add less value than selecting a missing enabler. Similar tradeoffs arise between combo pieces, mana sources, and early interaction. A useful recommendation therefore depends on what the pool already contains.

We study a draft assistant that starts from empirical card rankings and learns pool-dependent corrections. Candidate-to-pool attention identifies relevant previous selections, while conditional selection statistics provide a complementary signal from human drafts. Personal game results influence shared card representations and the weight assigned to the player's recorded choices. This approach obtains supervision from draft and game logs without requiring a manually specified rule for every card combination.

The paper describes the model trained on September 24, 2026 and examines three questions: how closely its choices match recorded picks, whether its scores respond to strategically relevant pool changes, and how observed game results vary across drafts and model versions. The evaluation separates these questions because agreement with a recorded choice, sensitivity to context, and improved gameplay are distinct outcomes.

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2. Previous Work

Contextual draft preferences. Bertram et al. (2021) formulate drafting as selecting an addition to an existing set and learn contextual preferences with a Siamese network and triplet loss. This establishes the central distinction between a card's standalone appeal and its suitability for the drafted pool. Our helper retains that contextual-choice setting but scores all offered candidates jointly under a pack-masked cross-entropy objective.

Bertram et al. (2024a) adapt InfoNCE to the same restricted-choice setting. They mask cards that were not offered and use row-wise cross-entropy rather than interpreting arbitrary cross-example pairings as negative evidence. Our pick loss follows the same conditional-choice formulation. Their implementation ranks cards by cosine similarity between card and pool encodings, with convolutions for the pool. Our implementation instead uses candidate-to-pool attention, seen-card history, and draft position, supplemented by an explicit prior and conditional selection associations.

Representations and new cards. The companion study by Bertram et al. (2024b) examines structured attributes, rules-text embeddings, card images, and statistical card-usage features. It evaluates transfer to card sets excluded from pick-model training and subsequent adaptation. It also includes pick-rate and win-rate information among its inputs, providing precedents for both semantic representations and statistical card features.

Our implementation uses learned identity embeddings, structured attributes, and frozen MiniLM text vectors, without card-image inputs. GIH win rate enters as a fixed additive prior whose ordering is enforced for an empty pool, rather than as a learned statistical input feature. For a new identity, we retain its actual features and rules text and estimate its identity embedding from text neighbors. Unlike the transfer experiments in Bertram et al. (2024b), our reported evaluation does not establish performance on an independently held-out card set. Availability of card-usage statistics must also be distinguished from availability of draft-training examples for those cards.

Contrastive-learning foundations. InfoNCE originates in contrastive predictive coding (van den Oord et al., 2018); CLIP applies contrastive matching to image–text representations (Radford et al., 2021). SigLIP replaces a globally normalized contrastive objective with a pairwise sigmoid loss (Zhai et al., 2023). These objectives provide the methodological background for contextual contrastive ranking. In our task, an unoffered card supplies no observed preference comparison. Restricting comparisons to the offered pack remains necessary regardless of whether a model uses a softmax or a pairwise formulation. Our auxiliary binary outcome loss predicts game results; it is not a SigLIP card–pool matching objective.

Outcome feedback and game-playing agents. Preference-based reinforcement learning studies policy learning from preference feedback (Wirth et al., 2017). Related card-game work trains policies spanning deck construction and battle in Legends of Code and Magic (Xi et al., 2023) and Hearthstone (Xiao et al., 2023). These provide broader context for learning deck quality from gameplay, but concern different games and training environments.

Our personal adaptation uses supervised pick imitation with outcome-based example weights, historical replay, reference-policy regularization, and an auxiliary loss using own-deck and observed opponent cards. It does not optimize a sequential drafting return through self-play, control battle actions, or learn from ranked trajectories. The present system combines these signals with contextual pick prediction for Powered Cube; its gameplay evaluation is observational.

3. Draft State and Data

3.1. Inputs and prediction target

At a decision, the state consists of the current pack X, drafted pool Y, previously observed but unpicked cards Z, pack and pick indices, and a flag indicating whether the observed history is complete. Pack indices run from zero to two, and pick indices from zero to fourteen. The policy produces a score for each offered card and masks all other cards. A softmax over offered scores supplies relative recommendation weights; these are not calibrated probabilities that a pick is optimal or that its eventual deck will win.

The operational goal is to assemble a playable deck, but the main supervised target is the card selected in a human log. This distinction limits the interpretation of imitation accuracy. A logged choice can be defensible without being uniquely best, and a good alternative can count as a mismatch.

3.2. Public draft corpus

The public training corpus consists of 17Lands Powered Cube 2025 draft records (17Lands, 2026). It contains 545 resolved card identities. Whole drafts are assigned by calendar date before sampling: training precedes November 13, 2025; validation runs from November 13 through November 15; and the later partition begins November 16. The retained training set contains 87,999 decisions from 2,000 drafts. Validation contains 13,200 decisions from 300 drafts.

Recorded pre-pick pool columns supply the drafted cards

directly. Preparation checks pool size, pack size, and membership of the recorded pick in the offered pack. Seen-card history is reconstructed only from observed packs. All rows in this release carry an incomplete-history flag; the model receives this limitation as an input rather than treating reconstructed observations as complete.

Evaluation uses the first 2,048 validation decisions, covering 47 drafts, including a partial final draft. Because validation informed representation development, these results measure performance on a reused development set. The later temporal partition is excluded from the reported evaluation.

3.3. Empirical card-quality prior

The prior uses a September 24, 2026 export of 17Lands Games in Hand win rates (GIH WR), a card-level statistic based on games in which a card appears in the player's hand. The export contains 539 numeric ratings and one missing rating for Lutri, the Spellchaser. The export's filters and observation window were not retained, limiting assessment of overlap with evaluation data.

The prior preserves the reported ordering without shrinking ratings toward the mean according to sample size. Consequently, cards with few observations can receive strong priors. These statistics capture associations with game outcomes, including differences in players and decks; they do not directly estimate the value of adding a card to a particular draft pool.

4. Scoring Model

4.1. Card representations and pool attention

Each card i has a 64-dimensional representation

$$e_i = E_i + g(f_i) + W t_i, \tag{1}$$

where E_i is a learned identity embedding, f_i contains structured card attributes, and t_i is a frozen 384-dimensional text embedding. Structured features include mana value, colors, card types, and other metadata. The feature projection is a two-layer network with a 128-unit hidden layer. The text encoder is `all-MiniLM-L6-v2` (Sentence Transformers, 2026); its projection into the card space is learned.

Encoder input combines mana cost, type line, rules text, and power/toughness. Card self-references are replaced with "this card." Long text is divided into overlapping token windows; the normalized mean of their embeddings avoids discarding the end of a long card. The generic encoder remains frozen during draft training. This gives the policy access to rules language, but does not implement a rules engine or prove target compatibility.

For each offered card, four-head attention uses its representation as the query and the drafted cards as keys and values. The candidate scorer receives the query, attended pool vector, mean pool representation, mean seen representation, learned pack/pick positions, normalized pool size, and history-completeness flag. A 128-unit hidden layer maps these inputs to a scalar $f_\theta(i, s)$. An empty pool uses a null attention key to keep the computation finite.

Candidate-specific attention can emphasize different pool cards for different choices. The mean summaries provide broader context. There is no explicit package-completion registry, mana-feasibility solver, or game simulator in the live scoring path.

4.2. Conditional association residual

A fixed matrix A_{ji} measures whether card i is selected more often when card j is already in the pool, conditional on i being offered. It is computed only from the public training split. For each candidate, selection rates with and without a pool card are smoothed toward its overall selection rate using 100 pseudo-observations. Their log-odds difference is clipped to $[-3, 3]$. Both groups require at least 50 offers; unsupported entries and the diagonal are zero.

Two learned scalar weights combine the mean association and strongest positive association:

$$a(i, Y) = \frac{\sum_j Y_j A_{ji}}{\max(1, \sum_j Y_j)}, \tag{2}$$

$$b(i, Y) = \max\big(0, \max_{j \in Y} A_{ji}\big), \tag{3}$$

$$r(i, Y) = w_1 a(i, Y) + w_2 b(i, Y). \tag{4}$$

The maximum is zero for an empty pool. The association matrix remains fixed during personal adaptation. Associations can capture compatible colors, archetypes, player preferences, and draft position as well as combos. They are statistical selection signals, not verified interaction labels.

4.3. Fixed prior and contextual correction

Let v_i be the exported GIH win rate. The prior is

$$P_i = \frac{\operatorname{logit}(v_i) - \operatorname{logit}(0.55)}{0.15}. \tag{5}$$

The centering rate is the median of numeric exported ratings. Missing ratings receive this median, giving a zero prior. The final score for an offered card is

$$S(i, s) = P_i + \mathbf{1}[|Y| > 0]\big(f_\theta(i, s) + r(i, Y)\big). \tag{6}$$

The prior is fixed during training. At the first pick, all learned corrections are gated off, so the model follows the exported GIH ordering exactly, with no specified strategic tie-break for equal ratings. Later picks combine the same prior with learned context. The scale 0.15 controls the prior's logit magnitude; it is not a win-probability calibration temperature.

5. Training and Personal Game Feedback

5.1. Training procedure

Initialization reuses learned card representations, text projections, and attention parameters from public pick training. The final candidate-scoring layer and the two association weights are reset to zero. This makes the initial scores exactly the GIH prior while retaining representations from which contextual corrections can be learned.

Context learning uses 1,000 AdamW steps, learning rate 3×10^{-4}, batch size 128, weight decay 0.01, and seed 924. Half of each batch is sampled uniformly from public training picks. Half is sampled from the upper quintile of the strongest positive chosen-card/pool association. These associations define sampling strata; the label remains the recorded human pick. Gradient norms are clipped at one.

Personal adaptation consists of a 100-step update using 135 picks and 62 games, followed by the final 100-step continuation using all 180 picks and all 67 games. Both adaptations use learning rate 10^{-5} and seed 922. The final continuation starts from the already adapted weights; examples in the earlier stage therefore receive additional exposure. Pick-prediction and contextual diagnostics evaluate the final model after the prescribed number of steps. All personal records are used in the final update, leaving no personal holdout within this training collection.

5.2. Personal records and draft linkage

The personal pick corpus contains four complete drafts of 45 picks each. Deduplication by draft identifier, pack index, and pick index removes 37 repeated log entries, retaining 180 distinct decisions. Repeated records must agree on the offered pack and chosen card.

All 67 personal training games are retained for the auxiliary outcome objective, including wins and losses. To weight personal picks, a game is associated with a logged draft only when its resolved nonbasic deck-card set contains at least 15 identities and is a subset of exactly one completed draft pool. This is an inferred link, not an explicit Arena draft identifier. Twenty-five games meet this matching criterion across the four drafts; the other 42 can still inform the auxiliary outcome objective without weighting a draft's picks.

For draft d with W_d matched wins among N_d matched games, its pick-example weight is

$$q_d = 0.5 + \frac{W_d + 2}{N_d + 4}. \tag{7}$$

The smoothing keeps small samples from dominating. Losing drafts retain positive imitation weight. The target remains the player's chosen card; a loss does not label every card in that draft as bad.

5.3. Learning from own and opponent cards

The auxiliary task predicts the player's game result from two separate card sets: the player's recorded nonbasic maindeck identities and observed nonbasic opponent identities. Basic lands are excluded and duplicate identities are collapsed. The opponent set is partial public information, not a recovered deck list or proof that every observed card was played.

For a binary card-presence vector x, let $n_x = \sum_i x_i$ and define

$$\mu(x) = \frac{\sum_i x_i e_i}{\max(1, n_x)}, \tag{8}$$

$$\nu(x) = \frac{(\sum_i x_i e_i)^{\odot 2} - \sum_i x_i e_i^{\odot 2}}{\max(1, n_x(n_x - 1))}. \tag{9}$$

Here $\odot 2$ denotes elementwise squaring. The second vector averages elementwise products of distinct card representations. It is a compact pair summary, not a separate learned coefficient for every named pair. Own and opponent means and pair summaries occupy distinct feature blocks; the calculation never treats cards on opposing sides as members of the same combo.

Five covariates record opponent observation count, game duration in turns, play/draw status, and separate counts of unresolved own and opponent identities. Four game records contain unresolved observations. These games remain in training; unknown identities are omitted from card vectors and represented by the count covariates. Four 64-dimensional card blocks plus five covariates yield 261 features.

A linear sigmoid head is first fitted for 200 Adam steps at learning rate 0.003 with squared-weight penalty 0.1, using frozen representations. The head is then frozen while the shared policy representations adapt. Each adaptation step samples 32 personal picks and 64 historical training picks and evaluates the auxiliary loss on all games in that stage's training set. The joint objective is

$$\mathcal{L} = \mathcal{L}_{\text{personal},q} + \mathcal{L}_{\text{historical}} + 0.5\, D_{\text{KL}}(\pi_{\text{ref}} \| \pi_\theta) + 0.2\, \mathcal{L}_{\text{outcome}}. \tag{10}$$

The first two terms are pick cross-entropies. The frozen reference is the policy at the start of the adaptation stage, and its KL term is evaluated on historical picks. Outcome loss is binary cross-entropy.

Thus wins, losses, and opponent observations can affect recommendations through shared representations, while matched results also weight personal choices. The outcome head influences the shared representations during training rather than adding a direct bonus to recommendation scores.

Visibility, player skill, matchup, and game duration confound these observations; the covariates do not remove those biases.

6. Live Recommendation

The assistant reads Arena's local event log, reconstructs the drafted pool and observed packs, and ranks all offered cards using Equation 6. It clears state at a new draft and checks that pool size matches draft position. On restart, it restores the active draft before processing new decisions. Recommendations include relative softmax weights and an indication of cards whose identity embeddings have been estimated.

For a card absent from the trained identity vocabulary, the assistant retains its structured attributes and locally encoded rules text. It estimates the identity embedding from eight trained text neighbors, sets its association entries to zero, and assigns a neutral prior if no rating is available. This procedure extends card coverage, although performance on independently held-out card sets has not been measured. Hardware settings, model selection, and launch commands are provided in Appendix A.

7. Evaluation

7.1. Study design

The evaluation combines recorded-pick agreement, nine paired contextual diagnostics, and retrospective game results from one player. The pick and diagnostic evaluations use the final September 24 model. The game analysis includes drafts without the helper and drafts associated with successive model versions. It is observational: neither model assignment nor recommendation adherence was controlled.

Public validation decisions measure imitation of human choices. The 180 personal picks and 67 games used for adaptation measure training fit. Separately, the game collection contains 55 additional outcomes absent from the training set. These additional records support a descriptive account of subsequent recorded results, but were not a prospectively specified test set. Appendix A documents data linkage, metric reproduction, and the handling of incomplete records.

7.2. Pick agreement and outcome fit

The final model matches 1,138 of 2,048 historical validation choices and 106 of 180 personal training choices (Table 1). The personal figure measures fit, not generalization to a new draft. The historical subset contains correlated picks from 47 draft identities and has been reused during development. We therefore do not present a pick-level confidence interval that would imply 2,048 independent drafting trials.

Table 1. Recorded-pick agreement for the final model. Historical results use a reused validation subset; personal results measure training fit. NLL is mean negative log likelihood of the recorded choice.

Data	Matches	Top-1	NLL
Historical validation	1138/2048	55.57%	1.2141
Personal training	106/180	58.89%	1.2189

Table 2. Auxiliary game-result prediction on the 67 training games. Log loss and Brier score measure prediction fit, not the win rate of following recommendations.

Scope	Log loss	Brier
67 training games	0.477294	0.148756

Table 2 reports game-level binary log loss and Brier score, the mean squared difference between predicted probability and binary result. The low-dimensional outcome fit uses only 67 games and has no final holdout. It cannot demonstrate calibrated prediction on future opponents, causal card value, or a higher win rate from using the helper. Current aggregate ratings also lack a documented collection window, so their independence from evaluation data cannot be certified.

7.3. Game outcomes by draft

The game collection contains 122 distinct Powered Cube games with 69 wins (56.6%). All records identify the same September 8, 2026 event. Seventeen complete draft histories can be uniquely matched to 108 of these games using the deck-subset criterion described above. The other 14 games have no unique match to a complete draft. Five additional complete draft histories have no matched outcomes and are excluded.

Figure 1 presents the outcome of each matched draft. The six drafts without the helper produced 17 wins in 35 games (48.6%). Drafts 7–9 are associated with the September 21 attention model, draft 10 with the September 22 personal adaptation, and drafts 11–17 with the final September 24 model. The final-model group produced 31 wins in 48 games (64.6%), with individual draft records ranging from 1/4 to 7/8. These aggregate rates weight games equally; each plotted point is a separate draft's rate.

The final-model group contains 315 recorded picks. Recorded choices match the first-ranked recommendation in 200 cases, or 179 of 294 (60.9%) after excluding forced final picks. However, the recommendation logs can include historical replay and do not consistently establish whether advice preceded the choice. Model association therefore identifies the configuration recorded for a draft, rather than proving that the player followed its recommendations.

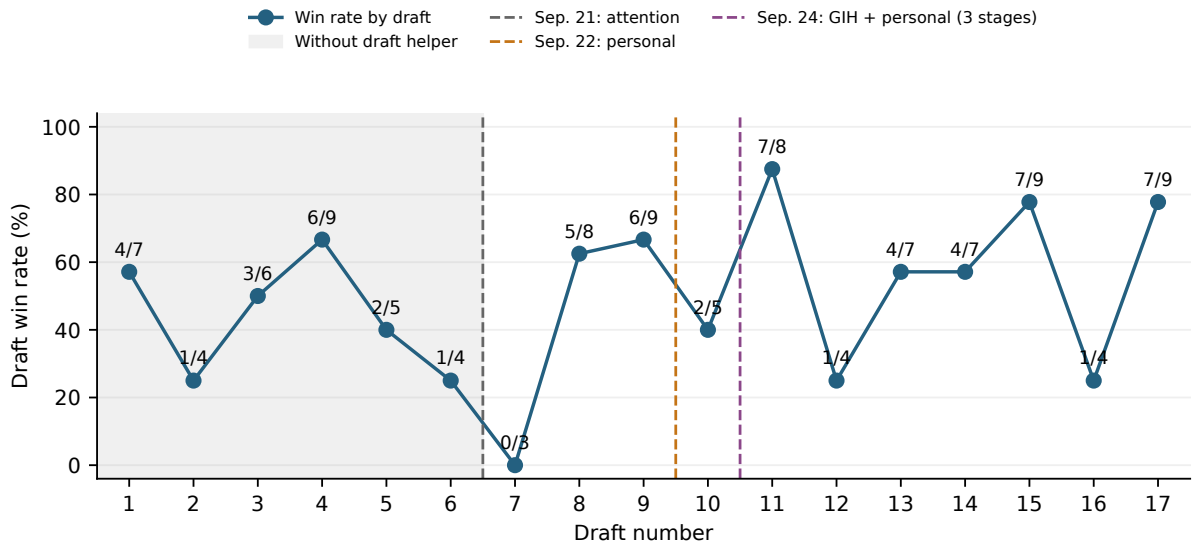


Figure 1. Win rate by draft. Each point shows wins divided by recorded games; labels give the corresponding counts. Shading identifies six drafts without the helper. Dashed lines identify model changes associated with training on September 21, 22, and 24. Each line precedes the first draft recorded with that model; it does not locate the exact training time. The three September 24 training stages share one boundary. Drafts follow reconstructed log order (Appendix B).

The higher observed win rate with the final model is consistent with improved performance, but the design cannot attribute the difference to the assistant. All drafts involve one player, with uncontrolled opponents, deck construction, and gameplay decisions. Games within a draft are correlated, and missing records may affect the composition of the sample. The small number of drafts also makes the aggregate sensitive to individual results. We report descriptive counts rather than an independent-game significance test. Log-date summaries for the full 122-game collection appear in Appendix B.

7.4. Training-data volume

Personal adaptation progresses from 135 picks and 62 games to 180 picks and 67 games. The earlier GIH-context stage uses public draft choices without personal examples. Only the final stage has an associated additional game cohort: all seven drafts in that group correspond to the same 180-pick/67-game model.

These observations do not identify a relationship between training-set size and win rate. Training-data volume is confounded with additional optimization and model changes, and there are no comparable game cohorts for the two preceding stages. The 50.7% win rate in the 67-game training collection is not a zero-personal-data baseline. A data-volume study would require models trained on different-sized subsets and evaluated under comparable gameplay conditions. Appendix B lists the available stages and outcome coverage.

Table 3. Contextual diagnostics for the final model. Each row changes one pool card while holding the pack fixed. The proposed sign describes the expected direction of the score-margin change.

Probe	Proposed sign	Margin change
Flash / Wurm	+	+0.140
Reanimate / target	+	-0.109
Entomb / setup	+	+0.127
Breach / Freeze	+	+0.261
Persist / legality	-	+0.391
Green fixing	+	+1.579
Control curve	+	+1.734
Channel / payoff	+	+0.745
Saga / Orb	+	+0.461

7.5. Contextual development probes

Eighteen authored development states form nine control/treatment pairs. Each pair holds the pack fixed and changes one pool card. For target a and reference b, we report

$$\Delta = \big[S(a, s_1) - S(b, s_1)\big] - \big[S(a, s_0) - S(b, s_0)\big]. \quad (11)$$

The fixed GIH prior cancels in this difference. It therefore measures a response of the learned contextual terms, even when the highest-ranked card remains unchanged. Probe histories are empty and flagged incomplete.

Seven comparisons follow the proposed direction (Table 3). Only the Channel comparison changes the first-ranked card: Llanowar Elves leads in the control pool, and Channel leads when its payoff is present. The other eight retain the same top choice across pools.

For the Breach comparison, adding Lion's Eye Diamond raises Brain Freeze's margin over Remand by 0.261, but Brain Freeze already leads in the control. Entomb similarly gains against Griselbrand when a reanimation target is added, while remaining first in both states. These examples distinguish sensitivity in the scores from a change in the recommended pick.

Two comparisons expose remaining weaknesses. Adding a target reduces Reanimate's margin over Griselbrand by 0.109, although Reanimate remains first. Replacing Archon with Titania increases Persist's margin over Reanimate by 0.391, contrary to the proposed compatibility response. Reanimate still outranks Persist in both states. These examples show why a plausible top pick does not establish correct reasoning about the pool.

The diagnostic cases were developed with AI assistance and reviewed informally by the player. The review allowed multiple acceptable choices and did not establish a complete independent label set. In particular, a within-state preference does not necessarily validate a proposed change in relative scores across states. Seven of nine therefore describes agreement with diagnostic expectations, not expert-rated pick accuracy or performance across all combo families. The cases were inspected during development and are not a held-out benchmark.

8. Discussion and Conclusion

The assistant combines a stable card-quality prior with learned adjustments for the developing pool. Attention, rules-text features, and conditional selection associations provide complementary ways to represent interactions. Personal outcome feedback supplies an additional learning signal without requiring card-by-card strategic labels. The contextual diagnostics show that these mechanisms produce measurable responses, but the reanimation failures demonstrate that they do not reliably enforce target compatibility or identify every missing role in a deck.

Several limitations restrict the conclusions. The public validation subset has been reused during model development, the personal adaptation uses only four drafts and 67 games, and the final training stage has no personal holdout. Opponent observations are selective, game-to-draft links are inferred, and the GIH export lacks documented filters. The gameplay study involves one player and provides neither randomized model assignment nor reliable measurement of recommendation adherence.

The observed results support further study of contextual drafting with personal feedback. They do not yet establish that the assistant improves win rate or that adding training games yields a predictable gain. Independent contextual evaluation and prospective trials across more drafts would help determine whether the measured score responses translate into stronger completed decks and gameplay.

Data Availability

The public draft source is 17Lands, attributed under CC BY 4.0 in the dataset manifest. Personal logs contain private game and draft identifiers and are not publicly released with this manuscript. Reported results use aggregate statistics and card-level diagnostics. Appendix A describes the model, data dependencies, and scripts required to reproduce the analysis.

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A. Implementation and Reproducibility

The evaluated model is the September 24, 2026 all-games checkpoint. Training used the Apple GPU through PyTorch MPS. The repository's `mlx` option is an alias for this backend, not a native MLX implementation. An explicit checkpoint argument selects the evaluated model; the unqualified `--model updated` option points to an earlier model.

Pick metrics were recomputed on CPU and matched the saved MPS results in top-choice counts and within 10^{-5} in negative log likelihood. Verification also checked all 539 numeric prior values and first-pick scores. Auxiliary outcome metrics are taken from the saved training report; the outcome head was not re-evaluated during manuscript preparation. Input and source hashes identify the evaluated artifacts.

All paths below are relative to the workspace unless stated otherwise. The final checkpoint SHA-256 is `420c87f9c47e7205925f5f0e47ffeb8a3600d1c24666c2e5913b9da884655f06`. This hash identifies the model used for pick prediction and contextual diagnostics.

From `mtga_cube_draft/`, launch the evaluated model with:

```
.venv/bin/python scripts/mtga_log_assistant.py \
  --model updated --device mlx --top-k 15 --from-start \
  --checkpoint reports/gih_all_games_2026_09_24/candidate.pt
```

The command ranks every offered card and follows new log events after restoring the latest draft. The process must remain active to provide live recommendations.

From the workspace root, regenerate paper evidence:

```
mtga_cube_draft/.venv/bin/python paper/build_evaluation.py
```

The script validates linked final-training input hashes, reloads the final policy, recomputes historical validation and personal training agreement, checks the fixed prior, and scores only the eighteen authored development states. It writes tables, per-pair scores, and provenance under `paper/generated/current_model/`. It does not train or evaluate the reserved later-time partition or reserved authored cases. The saved auxiliary-game metrics are explicitly distinguished from fresh pick and probe inference.

To reproduce the descriptive game-outcome analysis from its separate frozen local snapshot, run:

```
mtga_cube_draft/.venv/bin/python paper/build_winrate_analysis.py
```

This writes aggregate tables, the per-draft figure, and provenance under `paper/generated/winrate/`. It requires Matplotlib and the local snapshot `runs/paper_winrate_2026_10_04/` under the project. The snapshot preserves allowlisted outcome records, decisions, recommendations, and checkpoint fingerprints. Additional earlier pick histories are frozen in `runs/paper_earlier_drafts_2026_10_13/`; their manifest records source and snapshot hashes. The first six plotted drafts form the group without the draft helper. The local snapshots include private match and draft identifiers for reproducibility; the manuscript publishes only aggregates. The collection manifest hashes the source tracker files and the frozen inputs. Reproduction does not reread mutable tracker logs or retrain any model.

Artifact	Path relative to `mtga_cube_draft/`
Final model and training report	`reports/gih_all_games_2026_09_24/`
GIH initialization and context protocol	`reports/gih_context_2026_09_24/protocol.json`
Pre-final adaptation protocol	`reports/gih_personal_2026_09_24/protocol.json`
Public draft release	`data/datasets/real_context_2026_09_21_v2/manifest.json`
Text-embedding provenance	`data/datasets/real_context_2026_09_21_v2/text_manifest.json`
Context model and associations	`experiments/real_context_2026_09_21/model.py` and `improve.py`
GIH context training	`scripts/train_gih_context.py`
Personal adaptation	`scripts/finetune_recent_games.py`
Live scoring adapter	`scripts/attention_live.py`
Frozen diagnostic states	`benchmarks/powered_cube_2026_09/cases.jsonl`
Player review notes	`benchmarks/powered_cube_2026_09/user_review_2026_09_21.md`

The paper's generated tables are sufficient to compile a portable copy. Re-running the evidence generator also requires local datasets, private personal records, and checkpoint dependencies. The saved training protocols identify the exact source versions used during fitting; fresh inference provenance records the code used for this revision. Full build instructions are in `paper/README.md`.

B. Game Records and Training-Stage Coverage

Game records were collected through October 4, 2026 and deduplicated by match identifier and game number. Records with inconsistent results, event identities, or main decks were excluded; no such conflicts occurred in the analyzed collection. Additional archived pick histories provide the earlier drafts in Figure 1. The classification of the first six drafts as completed without the helper is supplied by the player.

Draft ordering preserves the order within each source's pick log. Earliest result-log dates order sequences that cannot be connected through overlapping draft histories. Because tracker software can replay earlier sessions, these dates are collection timestamps rather than verified game times. Checkpoint markers use recorded model hashes and the dates of their training runs. Multiple September 24 training stages precede the same next recorded draft, so the plot uses a single marker for that sequence of updates.

Table 4. Game outcomes by earliest tracker log date in 2026. The first two periods contain the 67 training games; the third contains 55 additional games. These collection periods are not randomized deployment groups.

Log-date period	Wins/games	Win rate
Sep. 8–20	20/40	50.0%
Sep. 21–23	14/27	51.9%
Sep. 24–28	35/55	63.6%

The training collection contains 34 wins in 67 games (50.7%); the additional collection contains 35 wins in 55 games (63.6%). The difference is 12.9 percentage points. This comparison uses all 122 games, whereas Figure 1 uses the 108 games uniquely matched to complete drafts. Neither comparison controls for player decisions, opponents, or missing records.

Table 5. Personal-data counts for the GIH-based model's training stages. Counts refer to distinct records within each stage, not cumulative training exposures. Only the final stage has associated additional game outcomes; missing entries do not denote zero wins.

Personal picks	Games used	Linked outcomes
0	0	Not attributable
135	62	Not attributable
180	67	31/48

The September 22 personal model shown in Figure 1 is a separate predecessor to the GIH-based training sequence. The three rows above refer to the September 24 context model, its first personal adaptation, and the final all-games update. The intermediate adaptation and final update share training examples, so their record counts must not be added together.